

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	Load and Electrical Equipment List
Document Number	EA-BFPCL-GPMC-GGPL-FEED-ELE-LST-001
Document Revision	R01
Document Status	Issued For Review
Originator	Noel Kari
Security Classification	Restricted
Issue Date	8 Jun 2022

Load and Electrical Equipment List

R01	08.06.22	Issued For Review	N.Kari	V.Ani	A. Duru	
Rev #	Issue Date	Status Description	Originator	Reviewer	Approver (Epic)	Approver (BFPCL)

Revision History

R01	08.06.22	Issued for Review	N.Kari	V.Ani			
Rev No	Date	Reason for Issue / Change	Prepared	Reviewed	TCE Approval	BFPCL Approval	Approver (BFPCL)

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.).
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

Restricted

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED-ELE-LST-001

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, GAS, PLANT, ROW, PTS

REFERENCES

Document No.	Description

Table of Contents

1. BBOU MANIFOLD
2. ELEPA MANIFOLD
3. AKABEL PIPELINE END FACILITIES

Epic Atlantic Limited | 12 Kenneth Agbakuru Street | Lekki Phase I, Lagos, NG | www.epicatlantic.com | info@epicatlantic.com 5 of 7

						DOCUMENT NUMBER			EA-BFPCL-GPMC-GGPL-FEED-ELE-LST-001				PROJECT:		REV	R02	DATE:
						DOCUMENT TITLE			ELECTRICAL LOAD LIST				ELEPA MANIFOLD				7 JUNE 2022
						A	B	C	D	E	F CONTINUOUS		G INTERMITTENT		H STANDBY		REMARKS
S/N	TAG NO.	LOAD DESCRIPTION	SUPPLY VOLTAGE	DUTY (C/I/S)	CRITICALITY (V/E/N)	LOAD	NOMINAL / MOTOR POWER	LOAD FACTOR A/B	EFF	POWER FACTOR	B*C/D	(B*C/D) X (tanθ)	B*C/D	(B*C/D) X (tanθ)	B*C/D	(B*C/D) X (tanθ)	
			(V)			(kW)	(kW)	(pu)	(pu)	(pu)	(kW)	(kVAr)	(kW)	(kVAr)	(kW)	(kVAr)	
1	1DB-01	Switchgear & Control Room Distribution Board	230	C	E	2.00	2.00	1.00	0.95	0.80	2.11	1.58	0.00	0.00	0.00	0.00	
2	1DB-02	Perimeter Lighting Distribution Board	230	C	E	0.70	0.70	1.00	1.00	0.80	0.70	0.53	0.00	0.00	0.00	0.00	
3	HOLD	400/230V AC UPS	400	C	V	5.00	5.00	1.00	0.95	0.80	5.26	3.95	0.00	0.00	0.00	0.00	
7	HOLD	Cathodic Protection Transformer	230	C	N	5.00	5.00	1.00	0.95	0.80	5.26	3.95	0.00	0.00	0.00	0.00	
8	HOLD	Corrosion Inhibitor Pump Motor A	230	C	N	1.00	1.00	1.00	0.90	0.80	1.11	0.83	0.00	0.00	0.00	0.00	
9	HOLD	Corrosion Inhibitor Pump Motor B	230	S	N	1.00	1.00	1.00	0.90	0.80	0.00	0.00	0.00	0.00	1.11	0.83	
10	HOLD	SPDC Power	230	C	E	3.00	3.00	1.00	1.00	0.80	3.00	2.25	0.00	0.00	0.00	0.00	
											17.44	13.08	0.00	0.00	1.11	0.83	

SUMMARY			
	kW	kVAr	kVA
Total Installed Load	18.55	13.92	23.19
Total Running Load (X%TC + Y%TI)	17.44	13.08	21.80
Peak Load (X%TC + Y%TI + Z%TS)	17.55	13.17	21.94

where:
 C = Continuous Load
 I = Intermittent Load
 S = Standby Load
 V = Vital
 E = Emergency
 N = Normal

TC = Total Continuous Load
 TI = Total Intermittent Load
 TS = Total Standby Load
 X% = 100%
 Y% = 30%
 Z% = 10%

NOTES:

- LOAD LIST WAS POPULATED BASED ON ASSUMED ESTIMATED LOADS
- LOAD ESTIMATE EXPECTED TO CHANGE CONSIDERABLE WHEN DETAILED P&ID/UFD IS ISSUED
- INSTRUMENTATION LOAD IS BASED ON ROUGH EST
- UPS SUPPLY SHALL BE VIA UPS DISTRIBUTION BOARD
- INSTRUMENT AIR PACKAGE BASED ON ROUGH EST

brass fertilizer						DOCUMENT NUMBER			EA-BFPCL-GPMC-GGPL-FEED-ELE-LST-001			PROJECT:		REV	R02	DATE:		
						DOCUMENT TITLE			ELECTRICAL LOAD LIST			AKABEL PIPELINE END FACILITIES				7 JUNE 2022		
						A	B	C	D	E	F		G		H			
S/N	TAG NO.	LOAD DESCRIPTION	SUPPLY VOLTAGE	DUTY (C/I/S)	CRITICALITY (V/E/N)	LOAD	NOMINAL /MOTOR POWER	LOAD FACTOR A/B	EFF	POWER FACTOR	B*C/D	(B*C/D) X (tanθ)	B*C/D	(B*C/D) X (tanθ)	B*C/D	(B*C/D) X (tanθ)	REMARKS	
			(V)			(kW)	(kW)	(pu)	(pu)	(pu)	(kW)	(kVAr)	(kW)	(kVAr)	(kW)	(kVAr)		
1	1DB-01	Switchgear &Control Room Distribution Board	230	C	E	2.00	2.00	1.00	0.95	0.80	2.11	1.58	0.00	0.00	0.00	0.00		
2	1DB-02	Perimeter Lighting Distribution Board	230	C	E	0.70	0.70	1.00	1.00	0.80	0.70	0.53	0.00	0.00	0.00	0.00		
3	HOLD	400/230V AC UPS	230	C	V	5.00	5.00	1.00	0.95	0.80	5.26	3.95	0.00	0.00	0.00	0.00		
7	HOLD	Cathodic Protection Transformer	230	C	N	5.00	5.00	1.00	0.95	0.80	5.26	3.95	0.00	0.00	0.00	0.00		
												13.33	10.00	0.00	0.00	0.00	0.00	

SUMMARY			
	kW	kVAr	kVA
Total Installed Load	13.33	10.00	16.66
Total Running Load (X%TC + Y%TI)	13.33	10.00	16.66
Peak Load (X%TC + Y%TI + Z%TS)	13.33	10.00	16.66

Where:
C = Continous Load
I = Intermittent Load
S = Standby Load
V = Vital
E = Emmergency
N = Normal

TC = Total Continous Load
TI = Total Intermittent Load
TS = Total Standby Load

X% = 100%
Y% = 30%
Z% = 10%

NOTES:

1. LOAD LIST WAS POPULATED BASED ON ASSUMED ESTIMATED LOADS	
2. LOAD ESTIMATE EXPECTED TO CHANGE CONSIDERABLE WHEN DETAILED P&ID/UPD IS ISSUED	3. INSTRUMENTATION LOAD IS BASED ON ROUGH ESTIMATE
4. UPS SUPPLY SHALL BE VIA UPS DISTRIBUTION BOARD	5. INSTRUMENT AIR PACKAGE BASED ON ROUGH ESTIMATE